

Material Traceability Report for Selective Laser Melting of 316L Stainless Steel Flat Washers

Report ID: MTR-SLM-316L-2023-10-B4 **Date of Issue:** 2023-10-20 **Prepared By:** Quality Assurance Department, Advanced Manufacturing Division **Approved By:** Dr. A. Richardson, Quality Manager

Executive Summary

This document provides full material traceability for production lot B4 of 316L stainless steel flat washers manufactured via Selective Laser Melting (SLM). The batch consisted of 33 finished washers with a total mass of 0.61 kg, produced during the October 2023 manufacturing cycle. All materials were tracked from receipt through powder atomization, SLM processing, and final disposition, with complete mass balance reconciliation and chain of custody records. The report verifies compliance with internal quality standards and material handling procedures.

1.0 Introduction

This traceability report documents the material flow for a specific production run of 316L stainless steel flat washers manufactured using Selective Laser Melting technology. The purpose is to maintain accurate records for quality assurance, regulatory compliance, and supply chain transparency. All mass measurements were performed using calibrated equipment, and materials were handled according to established procedures.

Manufacturing Dates: - Powder Atomization: 2023-10-16 - SLM Processing: 2023-10-17 - Final Inspection: 2023-10-18

Facility: Building 4, Additive Manufacturing Center **Equipment Used:** - Water Atomization Unit WA-07 - SLM Machine SLM-280HL Serial #2844 - Analytical Balance AB-5000 (Calibration Cert #CAL-2023-089)

2.0 Process Overview

The manufacturing process involves two primary stages: powder production via water atomization and component fabrication via Selective Laser Melting.

Water Atomization: Stainless steel feedstock is melted and atomized using high-pressure water jets to produce fine powder for SLM processing. Process water serves as both cooling and atomization media.

Selective Laser Melting: Metal powder is deposited in thin layers and selectively melted by a laser beam under inert argon atmosphere to build components layer-by-layer. Unused powder is recovered for reuse or recycling.

3.0 Material Receiving and Verification

All incoming materials were verified against purchase specifications and assigned unique identification codes.

3.1 Raw Material Receipt

Material Description	Supplier Lot Number	Internal Lot ID	Quantity Received (kg)	Certificate of Analysis	Receiving Date	Verified By
X2CrNiMo1712 Stainless Steel (316L base)	SMT-316L-23098	RM-316L-231016	4.11	COA #2847 (Grade 316L confirmed)	2023-10-15	T. Davies
Process Water (Deionized)	N/A	WTR-231016	16.8	Water purity report #445	2023-10-15	T. Davies

Verification Notes: - Stainless steel composition verified per ASTM A276 - Water conductivity measured at 0.8 µS/cm (within spec ≤1.0 µS/cm) - All materials released to production after QA approval

4.0 Powder Atomization Process

Water atomization was conducted on 2023-10-16 using WA-07 unit. Process parameters and material flows are documented below.

4.1 Process Parameters

- **Energy Consumption Rate:** 2.23 MJ per kg of stainless steel processed
- **Material Processed:** 4.11 kg of X2CrNiMo1712 stainless steel
- **Process Water Input:** 16.8 kg
- **Operating Duration:** 2.5 hours
- **Operator:** K. Patel

4.2 Material Outputs from Atomization

Material Description	Output Lot ID	Quantity (kg)	Disposition	Notes
316L Stainless Steel Powder	POW-316L-231016	4.11	Transferred to SLM department	Particle size distribution: 15-45 µm
Recovered Process Water	WTR-REC-231016	16.4	Returned to water treatment system	Filtration efficiency: 97.6%
Solid Waste (Atomization Sludge)	WST-ATM-231016	0.41	Landfill disposal	Hazardous waste characterization: Non-hazardous

Quality Checks: Powder oxygen content: 210 ppm (within spec ≤250 ppm), Flow rate: 28 s/50g (within spec 25-35 s/50g)

5.0 Selective Laser Melting Manufacturing

SLM processing was completed on 2023-10-17 using 316L powder from lot POW-316L-231016.

5.1 Process Parameters

Energy Consumption: - Processing time per working cycle: 13.38 hours - Nominal power consumption with laser active: 5.5 kW - Nominal power consumption with laser inactive: 3.5 kW

Argon Shielding Gas: - Initial chamber purge volume: 700 liters - Additional volume per component: 54 liters - Number of components in build cycle: 33 pieces

Build Parameters: - Layer thickness: 30 µm - Laser power: 200 W - Scan speed: 800 mm/s - Build plate temperature: 80°C

5.2 Material Inputs to SLM

Material Description	Source Lot ID	Quantity (kg)
316L Stainless Steel Powder	POW-316L-231016	4.11
<i>Note: Argon gas consumption calculated based on flow meter readings and process parameters</i>		

5.3 Material Outputs from SLM

Material Description	Output ID	Quantity	Disposition	Notes
Finished Flat Washers (316L)	WAS-316L-231017	0.61 kg (33 units)	Quality approved, released to inventory	Dimensional inspection: All within tolerance
Reusable 316L Powder	POW-REC-231017	2.94 kg	Returned to powder handling system	Sieve analysis: 98.2% within 15-45 µm
Powder for Remelting	POW-RMT-231017	0.15 kg	Returned to atomization department	Contamination level: <0.1%
Non-recyclable Powder Waste	WST-PWD-231017	0.01 kg	Landfill disposal	Particle agglomeration noted

Process Efficiency: Powder reuse rate: 95.4% (industry benchmark: 90-97%)

6.0 Output and Waste Management

6.1 Product Output

Product Description	Part Number	Quantity	Total Mass (kg)	Final Disposition
316L Stainless Steel Flat Washers	FLW-316L-10M	33 units	0.61	Packaged and shipped to customer CO-2849

6.2 Recovered Materials

Material Description	Quantity (kg)	Processing Method	Next Use
Recovered Process Water	16.4	Filtration and treatment	Reuse in atomization process
Reusable 316L Powder	2.94	Sieving and characterization	Next SLM build cycle
Powder for Remelting	0.15	Collection and packaging	Water atomization feedstock

6.3 Waste Streams

Waste Description	Quantity (kg)	Disposal Method	Waste Code
Solid Waste (Atomization)	0.41	Secure landfill	NWML-184
Non-recyclable Powder	0.01	Secure landfill	NWML-185

Waste Management Compliance: All disposals documented per environmental permit EP-2023-44

7.0 Mass Balance Summary

The material mass balance verifies accountability across the manufacturing process.

7.1 Input Mass Summary

Material	Quantity (kg)
X2CrNiMo1712 Stainless Steel	4.11
Process Water	16.8
Total Input Mass	20.91

7.2 Output Mass Summary

Material	Quantity (kg)
Finished Washers	0.61
Reusable Powder	2.94
Powder for Remelting	0.15
Recovered Water	16.4
Solid Waste	0.41
Powder Waste	0.01
Total Output Mass	20.52

Mass Balance Variance: -0.39 kg (-1.9%)

Variance Analysis: The 0.39 kg variance is within acceptable limits ($\pm 2\%$) and attributed to measurement uncertainty and minor process losses including water evaporation during atomization. Last quarter average variance: -1.2%.

8.0 Chain of Custody

Complete material handling records ensure full traceability.

8.1 Material Transfer Log

Date	Material	From	To	Quantity	Transferred By	Verified By
2023-10-16	X2CrNiMo1712 Stainless Steel	Receiving	Atomization	4.11 kg	T. Davies	K. Patel
2023-10-16	316L Powder	Atomization	SLM	4.11 kg	K. Patel	M. Chen
2023-10-17	Finished Washers	SLM	Inspection	0.61 kg	M. Chen	A. Kumar
2023-10-17	Reusable Powder	SLM	Powder Storage	2.94 kg	M. Chen	A. Kumar
2023-10-18	Powder for Remelting	SLM	Atomization	0.15 kg	M. Chen	K. Patel

8.2 Quality Hold & Release

Material	Hold Date	Reason	Resolution	Released Date	Released By
316L Powder	2023-10-16	Routine sampling	Quality confirmed	2023-10-16	A. Kumar
Finished Washers	2023-10-17	Dimensional verification	All parts within spec	2023-10-18	A. Kumar

9.0 Quality Assurance and Compliance

9.1 Inspection Records

Raw Material Inspection: - Stainless steel: Composition verified per EN 10088-3 - Process water: Purity confirmed per internal specification QS-084

In-Process Inspection: - Powder quality: Sampled every 2 hours during atomization - SLM process: Layer continuity monitored via onboard sensors

Final Product Inspection: - 100% visual inspection completed - Dimensional checks on 10% sample: All within drawing tolerance - Surface roughness: Ra 12.3 µm (spec: Ra ≤15 µm)

9.2 Compliance Statements

This manufacturing lot complies with: - Internal Quality Standard QS-217 (Material Traceability) - Environmental Management Procedure EMP-033 (Waste Handling) - Industry Specification ASTM F3184-16 (Additive Manufacturing)

Documentation Retention: All records maintained for minimum 10 years per quality system requirements.

10.0 Conclusion

The material traceability for SLM production of 316L stainless steel flat washers (Lot B4) has been fully documented and verified. All materials were accounted for throughout the manufacturing process with acceptable mass balance variance. The 33 finished washers meet all quality specifications and have been released to inventory.

Report Closure Date: 2023-10-21 **Next Audit Schedule:** 2024-01-15

End of Report